Devanshu Bhargava

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github.com/devanshubhargava Portfolio

Summary

Backend-focused Software Engineer with 2+ years of experience building event-driven systems in Java, Spring Boot and Apache Kafka on an enterprise payment fraud detection platform. Builds for correctness under failure (idempotency, outbox, DLQs, circuit breakers, backpressure), proven with integration, load and chaos tests; deploys on AWS with Terraform.

Experience

Infinite Computer Solutions **Oct 2024 – Present**
Associate Software Engineer *Noida*

- **Backend Development:** Contribute to backend services of an enterprise **fraud detection platform** processing high-volume payment transactions in a **10+ member** team supporting real-time risk analysis.
- **Event Streaming Pipeline:** Implemented and maintained **Apache Kafka consumers** ingesting high-frequency payment events, with manual offset handling and retry logic.
- **Failure Handling & DLQ:** Added a **Dead Letter Queue (DLQ)** for fraud events failing after retry thresholds, improving fault tolerance and debugging of malformed messages.
- **Database Performance:** Helped resolve a persistence bottleneck with **Kafka consumer batch processing** and **PostgreSQL** bulk inserts, significantly improving write throughput.

Projects

NEXUS — Real-Time Fleet Tracking Platform | *Java 21, Kafka, TimescaleDB, Redis, Next.js, AWS*

- **Hand-Built Ingestion:** Plain kafka-clients pipeline committing offsets after the DB commit, and loading GPS events into **TimescaleDB** via COPY with idempotent writes; **k6**-tested on a 50K-vehicle model.
- **Fault Tolerance:** DLQ, **bisection recovery** for bad rows and partition-pause **backpressure**; crashes, replays and DB outages covered by Testcontainers and Toxiproxy tests.
- **Live Map:** Next.js + **MapLibre GL** map with Web Workers, fed by **Redis Pub/Sub** WebSocket fan-out; deployed on AWS ECS via Terraform with Grafana lag and freshness alerts.

CORTEX — Multi-Tenant GraphRAG Research Platform | *Python, LangGraph, pgvector, Neo4j, Bedrock*

- **Hybrid Retrieval:** Postgres full-text + **pgvector** HNSW, Reciprocal Rank Fusion, cross-encoder reranking and **Neo4j GraphRAG**; recall@k and MRR evaluated nightly in CI on a hand-labelled golden set.
- **Multi-Agent Orchestration: LangGraph** supervisor over Researcher, Analyst and Coder agents with loop guard and token budgets; cost-aware Ollama/Bedrock routing; sandboxed code execution.
- **Multi-Tenant Platform:** Tenant isolation with PostgreSQL **Row-Level Security**; SSE streaming from **Celery** workers to a Next.js UI with citation drill-down.

Technical Skills

Languages: Java 21, Python, TypeScript, JavaScript, SQL
Backend & Distributed Systems: Spring Boot, FastAPI, gRPC, REST, Microservices, Kafka, Redis, Celery, Resilience4j
Databases: PostgreSQL, TimescaleDB, pgvector, Neo4j
Generative AI & ML: RAG, LangGraph, LangChain, Amazon Bedrock, Ollama, scikit-learn, MLflow
Cloud & DevOps: AWS (ECS, RDS, ElastiCache, S3, Cognito), Terraform, Docker, Kubernetes, GitHub Actions
Frontend & Testing: React, Next.js, Testcontainers, JUnit 5, pytest, k6, OpenTelemetry, Prometheus, Grafana

Education

Rajiv Gandhi Proudyogiki Vishwavidyalaya **June 2020 – June 2024**
B.Tech, Computer Science Engineering | CGPA: 7.62 *Madhya Pradesh*

Achievements

Google Data Analytics Professional Certificate — Verify
Google Advanced Data Analytics Professional Certificate — Verify
Achieved Performer of the Month among 200+ employees — Verify